

Day 2 – AI / LLM Learning (Detailed Summary with Examples)

1. What an LLM Really Does

An LLM (Large Language Model) does not think or understand like a human. Its core job is very simple: it predicts the next token based on the tokens that came before it. It repeats this process again and again to build a full answer.

Example: If you type: “The capital of France is”, the model internally considers multiple possible next tokens like “Paris”, “Lyon”, or “Marseille” with different probabilities. Since “Paris” has the highest probability, it usually picks “Paris”, then continues predicting the next token such as “.” to complete the sentence.

2. Why This Feels Like Intelligence

The model has been trained on huge amounts of text and has learned patterns such as question–answer formats, code structures, explanations, and conversations. Because it predicts very good next tokens based on these patterns, the output often looks like real reasoning or understanding, even though it is actually advanced pattern prediction.

3. Why AI Can Be Confidently Wrong (Hallucination)

Since the model predicts what sounds likely instead of checking a real database of facts, it can sometimes produce answers that look confident but are wrong. For example, if you ask about the CEO of a company that does not exist, the model may still generate a realistic-sounding name because it has learned the pattern: “The CEO of X is Y.”

This is why AI output must always be reviewed and verified, especially for factual, legal, medical, or production code usage.

4. Temperature: Creativity vs Control

Temperature controls how random or creative the model is when choosing the next token from the probability list. At low temperature (for example 0.1–0.3), the model almost always picks the highest-probability token, making the output more deterministic and reliable. This is best for code, SQL, configuration files, and bug fixing.

At medium temperature (around 0.5–0.7), the model may sometimes pick the second or third best option, making the output more natural for explanations and tutorials. At high temperature (0.9 or above), the model explores lower-probability tokens, which increases creativity but also increases the risk of mistakes. This is useful for stories, brainstorming, and idea generation.

Practical rule: Use low temperature for development work to avoid broken or hallucinated code.

5. Context Window: The Model’s Working Memory

The context window is the amount of text (measured in tokens) the model can see at one time. It includes your messages, the model’s replies, system instructions, and any pasted code. You can think of it as the model’s short-term or working memory.

When the context window becomes full, the oldest information is pushed out. This means the model can forget earlier instructions, constraints, or parts of the conversation. This is why very long chats or pasting huge amounts of code can lead to worse or inconsistent answers.

Practical tips: Summarize long conversations, paste only relevant code, restate important constraints, and break big tasks into smaller steps.

6. Prompting: Steering the Probabilities

A prompt is not just a question; it is a way to guide the model's probability space. By giving clear roles, constraints, and output expectations, you push the model toward better and more useful answers.

Bad prompt example: "Fix this code." Better prompt example: "You are a senior backend developer. Fix this FastAPI code. Do not change the business logic. Only fix the bug and explain what you changed. Here is the code: ..."

The better prompt works because it clearly defines the role, scope, constraints, and expected output format, which reduces mistakes and hallucinations.

7. Key Takeaways from Day 2

- 1 LLMs predict the next token; they do not think or reason like humans.
- 2 What looks like intelligence comes from learning huge numbers of text patterns.
- 3 AI can hallucinate because it predicts what sounds likely, not what is guaranteed true.
- 4 Temperature controls randomness: low for reliability, high for creativity.
- 5 For coding and technical tasks, always prefer a low temperature.
- 6 The context window is limited working memory; when it is full, older info is forgotten.
- 7 Long chats and large code pastes can reduce answer quality if context is lost.
- 8 Good prompts guide the model with clear roles, constraints, and output format.